Small Moduli, Big Impact: Toward Communication-Efficient Lattice-Based OPRFs

Lena Heimberger

Monash University

1st of December, 2025

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  - verifiable Naor-Reingold PRFs and consistency proofs

Graz, Austr$_{al}$ia, Republic of

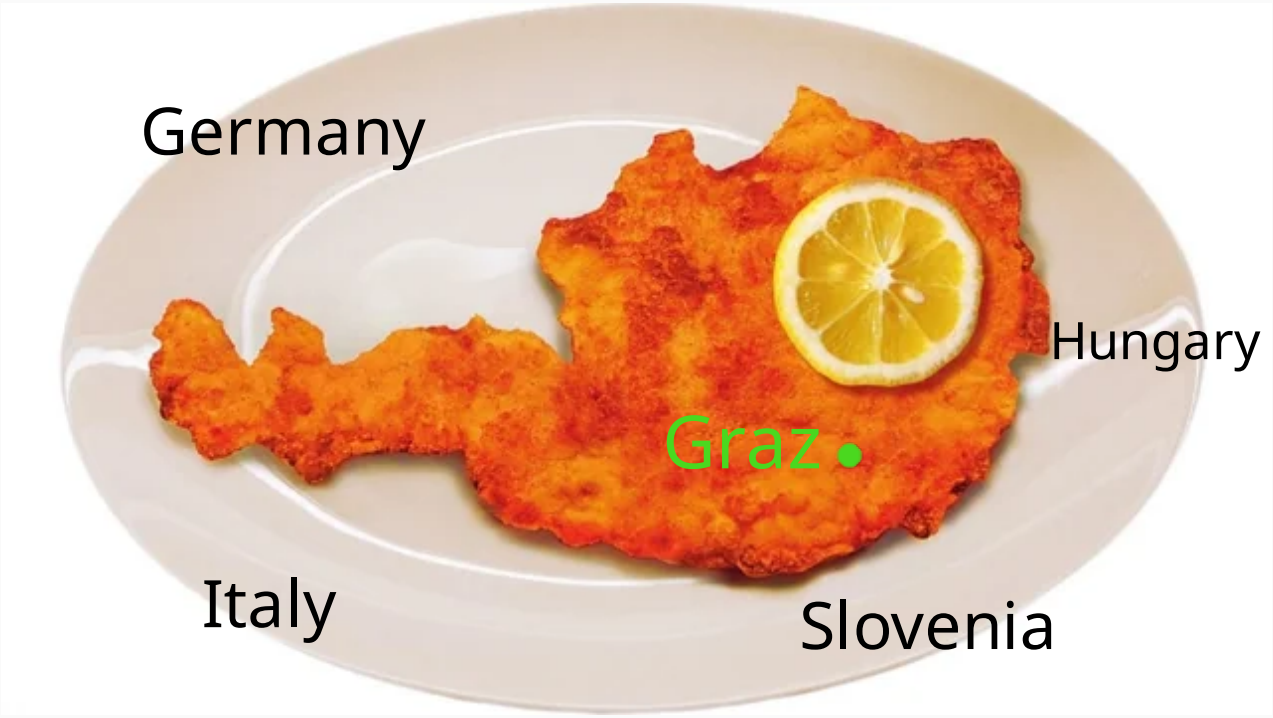

Figure 1: Schnitzel map from `https://www.reddit.com/r/mapporncirclejerk/comments/e3x9h4/map_of_austria_but_its_actually_a_schnitzel/`, modified by me.

Motivation for Oblivious Pseudorandom Functions

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Pseudorandom Function

A pseudorandom function (PRF) [GGM84, GGM86] is a deterministic and polynomial time function $F\colon \{0,1\}^k \times \{0,1\}^x \to \{0,1\}^y$ such that there is no probabilistic polynomial-time algorithm to distinguish any output y from a randomly chosen element from $\{0,1\}^y$.

Definitions

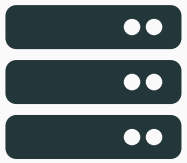
Client

input $c \in \mathcal{C}$, gets y

compute

$$y := F_k(c)$$

Server



key $k \in \mathcal{K}$

Oblivious Pseudorandom Function

An oblivious pseudorandom function (OPRF) [FIPR05] is a protocol between two parties. One party holds the secret key k and the other holds their secret input x. The OPRF privately realizes the joint computation outputting $F_k(x)$ for a PRF F to the party holding x, and nothing to the party holding k.

Additional Properties

Partial Obliviousness In a POPRF, the client submits a tag t in addition to the input x and gets the evaluation $y := F_k(t, x)$. t contains additional information, e.g. a validity period.

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Public Verifiability means we built a blind signature.

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Communication size. Elliptic curve OPRFs: 766 bits of communication. Best lattice-based: tens of kilobytes, in a semi-honest setting and with preprocessing.

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Trusted Setup. In addition to preprocessing, some OPRFs rely on a trusted setup.But: trust is a hard problem in itself, and sometimes a trusted setup just isn't available (e.g. for CSIDH).

OPAQUE

	Client (pwd)		**Serve**
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- Verifiability?

Private Set Intersection (PSI)

OPRFs are useful for unbalanced sets, where one party, typically the server in client-server protocols, has a significantly larger set than the other party.

Alice $(x_1, \ldots, x_m)$ **Bob** $(y_1, \ldots, y_n), k$

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- PSI can be a good base case for non-traditional OPRFs

Naor-Reingold and Lattices

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Good framework to study group actions!

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Multiplication is addition of exponents!

$$g^a g^b = g^{a+b} \text{ for some generator } g$$

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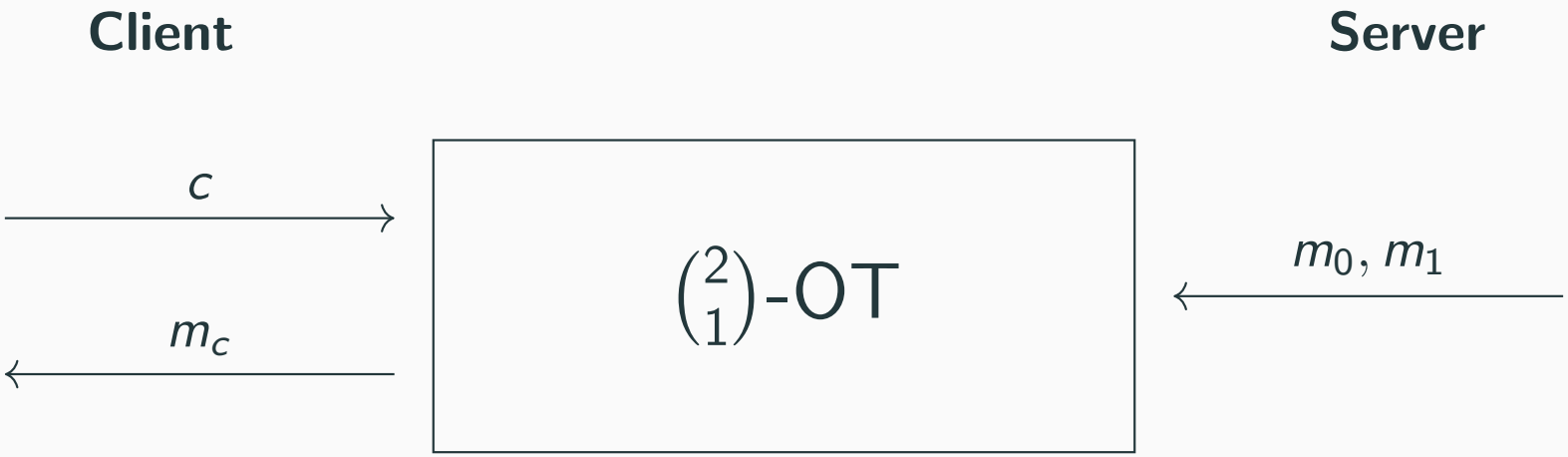
  - 2-component is evaluated similar to 257-component

Oblivious Evaluation

A little help from MPC friends



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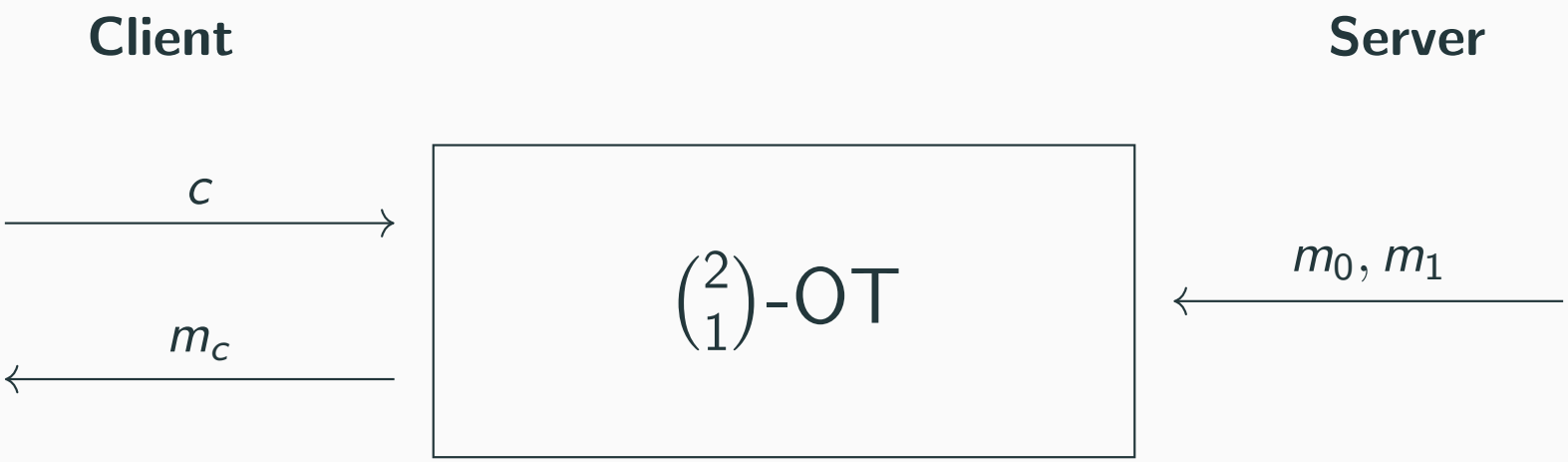

Extend with symmetric operations!

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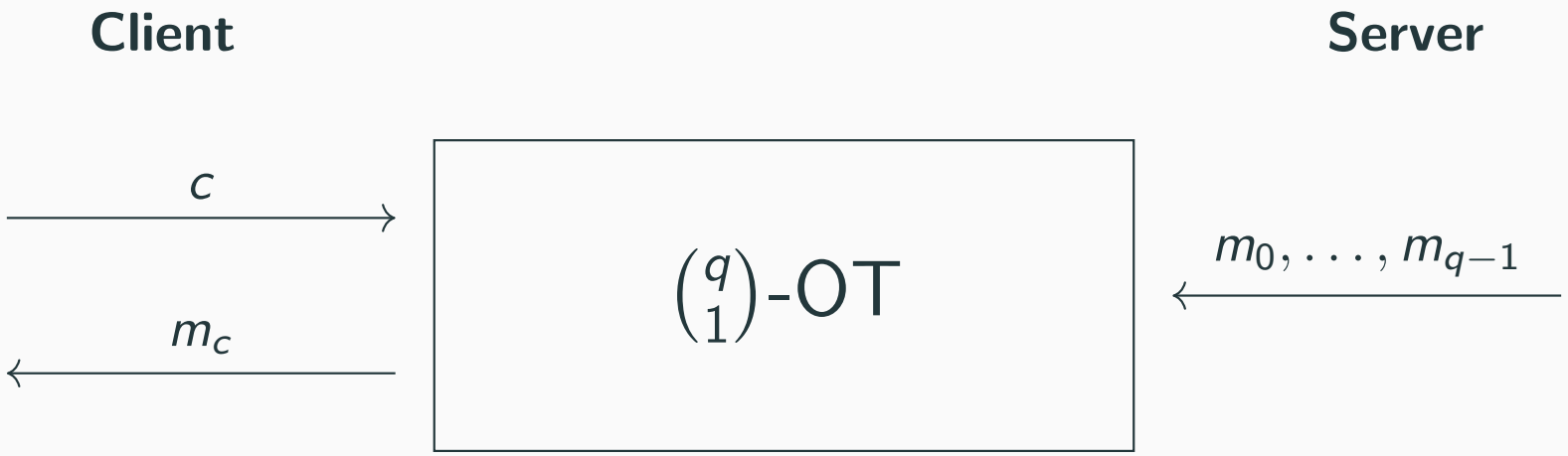

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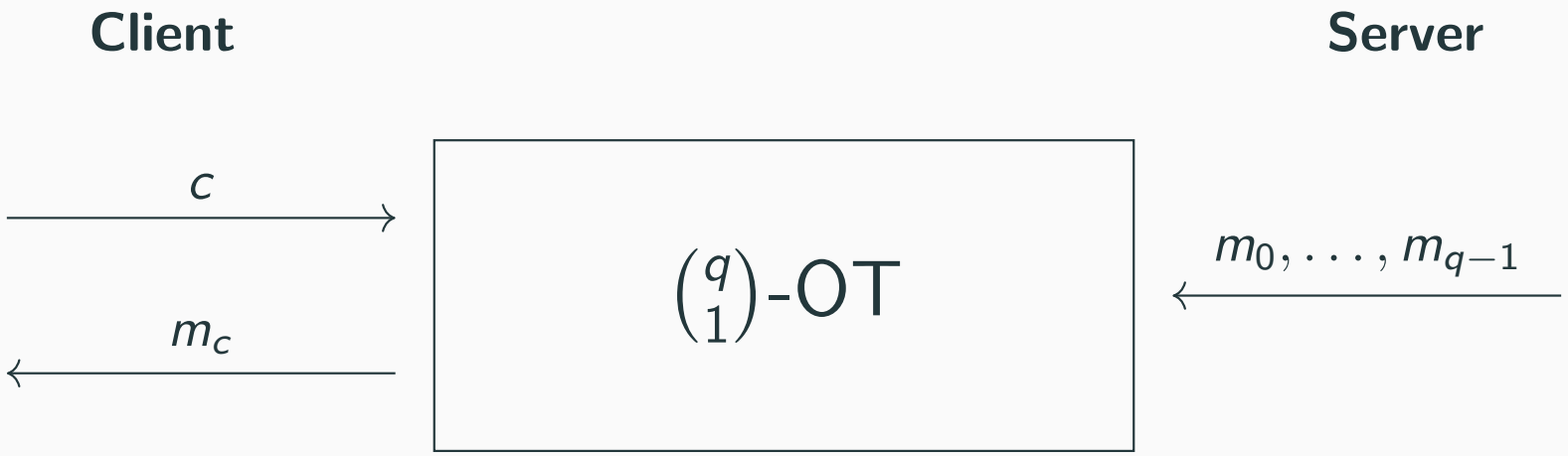

Generic construction from $\log q$ $\binom{2}{1}$-OTs !

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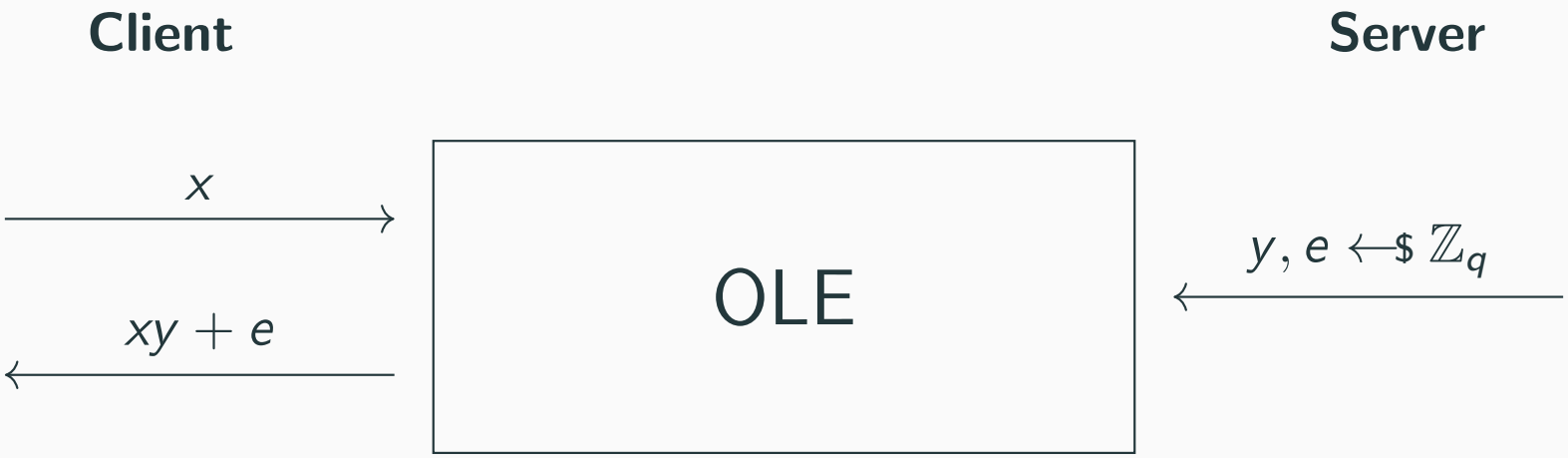

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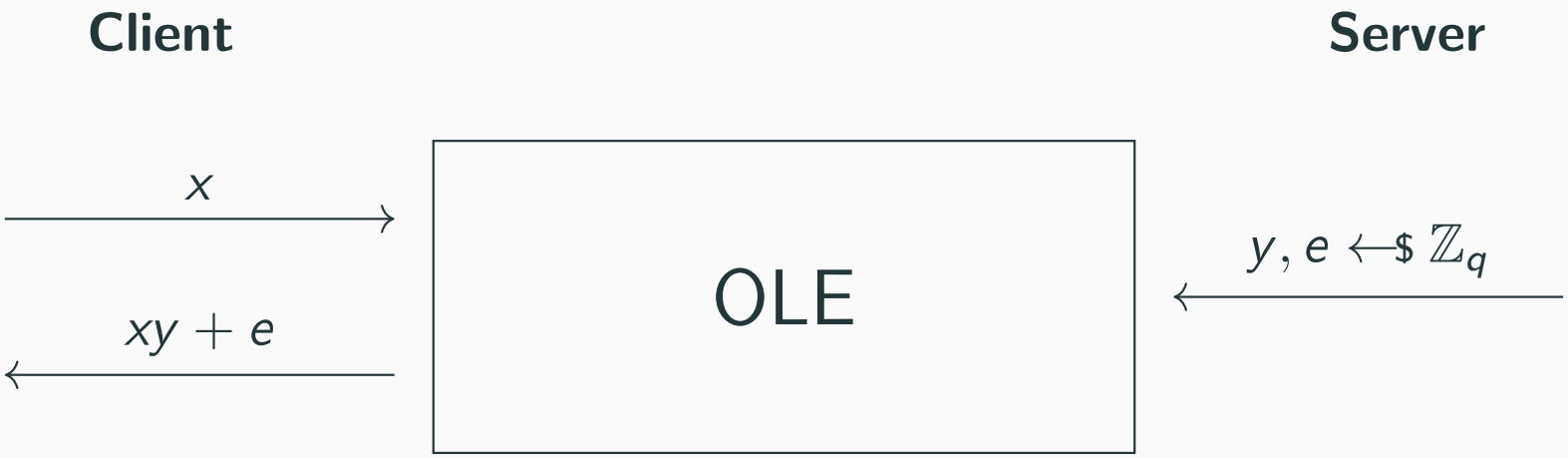

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- online phase takes less than a millisecond

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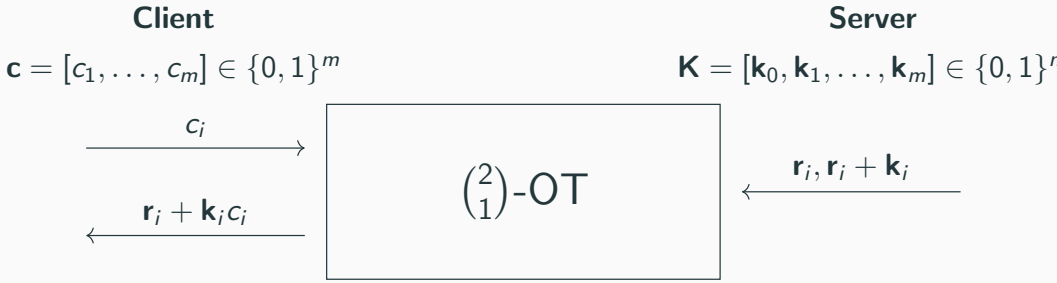
An OPRF from Spring



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$\xrightarrow{y_i}$ $\xleftarrow{\lfloor y_i - e_i \rceil_p \oplus d_i}$	$\binom{q}{1}$-OT	$\xleftarrow{\lfloor (q_i - e_i) \rceil_p \oplus d_i}$ $\forall q_i \in \mathbb{Z}_q$
	$\xleftarrow{\text{BCH}(\mathbf{d})}$	

I am very proud of it

- ✔ proof in the semi-honest model

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Solution? Verifiable variant of this OPRF. But Naor-Reingold keys are rather large which makes the verification less efficient.

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- The key observation is that the rounding result is deterministic and constant for fixed inputs.
- Add an offset Δ to the coefficient and leak the exact point where the rounding result flips from zero to one or vice versa.
- The attack is mitigated by enforcing consistency between the OLE output and the rounding input so the client cannot compute the offset anymore.

Observation on the modulus

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- ... combine both to mitigate attack?
- We suggested this!
  Still leak key offset, but cannot determine actual coefficient
- Probability of $\frac{1}{2}^n$ of guessing the correct key!
- Cool, but hacky: we leak a lot about the key. The original goal was designing lattice PRFs without ZK.

Let's work it out in the remix

- Leak everything but the MSB?

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- Lattice attacks?

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  Some problems to make it malicious secure.
- Parameter choice may be sub-optimal.
- What is the concrete best attack on tiny moduli?
- How to prove malicious security with that much leakage and interaction? UC does not work (blind computation).

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- Thank you, Sonja, Hannah, Sebastian, Reinhard, Shibam, Markus and Christian for your helpful comments on an earlier version of this lecture!
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Small Moduli, Big Impact: Toward Communication-Efficient Lattice-Based OPRFs

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Monash University

1st of December, 2025

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